

Shikang Wu

✉ wushikang@bytedance.com | 🌐 shikangwu.github.io | 🎓 Google Scholar | 🔗 LinkedIn

RESEARCH INTERESTS

Recommender systems, information retrieval, large-scale ranking models, multimodal recommendation, representation learning, and generative recommendation.

EMPLOYMENT HISTORY

ByteDance	Beijing, China
<i>Staff Machine Learning Scientist (3-1), Douyin-Search</i>	2026 – Present
- Leading a team to build next-generation personalized search engines and large-scale recommendation models for industrial search systems. - Current work spans generative foundation models for search, pre-training and post-training, long-sequence modeling, and scalable ranking architectures.	
<i>Senior Machine Learning Scientist (2-2), Douyin-Search</i>	2025 – 2026
Lead of the Large Recommendation Model (LRM) for Douyin Search, focusing on model scaling, sequential modeling, all-token design, multi-modal co-training, and next-generation search tools.	
<i>Senior Machine Learning Scientist (2-1), Douyin-Search</i>	2024 – 2025
- Lead of the Large Recommendation Model (LRM) for Douyin Search. - Focused on sparse activation scaling ($30\times$, using MoE, memory networks, etc.), sequence scaling ($20\times$), sparse scaling ($10\times$), and a multi-distribution learning all-token design.	
Baidu	Beijing, China
<i>Senior Machine Learning Scientist (T5, cross-level promotion), Baidu Search</i>	2023 – 2023
<i>Machine Learning Scientist (T3), Baidu Search</i>	2021 – 2023
Meituan	Beijing, China
<i>Machine Learning Scientist Intern, Recommendation</i>	Jun. 2020 – Sep. 2020

EDUCATION

Beijing Jiaotong University <i>M.E., School of Information Technology; advised by Zhihao Wu and Youfang Lin</i>	Sep. 2018 – Jun. 2021 Beijing, China
Beijing Jiaotong University <i>B.E., School of Information Technology</i>	Sep. 2014 – Jun. 2018 Beijing, China

PUBLICATIONS

MSN: A Memory-based Sparse Activation Scaling Framework for Large-scale Industrial Recommendation.

- Shikang Wu***, Hui Lu*, Jinqiu Jin*, Zheng Chai*, Shiyong Hong, Junjie Zhang, Shanlei Mu, Kaiyuan Ma, Tianyi Liu, Yuchao Zheng, Zhe Wang[†], Jingjian Lin. *KDD 2026*. [arXiv](#)

MDL: A Unified Multi-Distribution Learner in Large-scale Industrial Recommendation through Tokenization.

- Shanlei Mu*, Yuchen Jiang*, **Shikang Wu***, Shiyong Hong, Tianmu Sha, Junjie Zhang, Jie Zhu, Zhe Chen, Zhe Wang[†], Jingjian Lin. *KDD 2026*. [arXiv](#)

TokenMixer-Large: Scaling Up Large Ranking Models in Industrial Recommenders.

- Yuchen Jiang*, Jie Zhu*, Xintian Han*, Hui Lu*, Kunmin Bai*, Mingyu Yang*, **Shikang Wu***, Ruihao Zhang*, Wenlin Zhao*, Shipeng Bai, Sijin Zhou, Huizhi Yang, Tianyi Liu, Wenda Liu, Ziyang Gong, Haoran Ding, Zheng Chai, Deping Xie, Zhe Chen, Yuchao Zheng[†], Peng Xu. *KDD 2026*. [arXiv](#)

HyFormer: Revisiting the Roles of Sequence Modeling and Feature Interaction in CTR Prediction.

- Yunwen Huang*, Shiyong Hong*, Xijun Xiao*, Jinqiu Jin*, Xuanyuan Luo, Zhe Wang, Zheng Chai[†], **Shikang Wu[†]**, Yuchao Zheng, Jingjian Lin. *SIGIR 2026*. [arXiv](#)

LEMUR: Large scale End-to-end MULTimodal Recommendation.

- Xintian Han*, Honggang Chen*, Quan Lin*, Jingyue Gao*, Xiangyuan Ren*, Lifei Zhu, Zhisheng Ye, **Shikang Wu**, XiongHang Xie, Xiaochu Gan, Bingzheng Wei, Peng Xu, Zhe Wang, Yuchao Zheng[†], Jingjian Lin, Di Wu, Junfeng Ge. *CIKM 2026*, under review. [arXiv](#)

NOTES

* indicates equal contribution; [†] indicates corresponding author.